

Schedule 2024-25

(Provisional)

Week (Uni.)	Lecture	Practical
1 (27 th Jan.)	Lecture 1: Introduction Module overview Introduction.pptx Fundamentals of ML - Linear regression and gradient descent - Cost functions - Use sklearn LinearRegression class Linear Regression and Gradient Descent.pptx	Setting up and Running Jupyter Notebook - Getting ready for the module - using Jupyter Notebook on CSCT cloud-based jupyterhub server and Google Colab (The instructions on Colab to follow when using it) Setting up Jupyter Notebook.pdf Tutorial on Python, Numpy, Panda, Matplotlib and Scikit-Learn with Jupyter and Colab intro-to-libraries.ipynb (This is a reference notebook only rather than part of any practical worksheet - the appropriate methods are introduced when they are used in worksheets.) Practical 1: - Tutorials and exercises on linear regression with gradient descent and sklearn linear-regression-sklearn-gradient-decent.ipynb
2 (3rd Feb.)	Lecture 2: Fundamentals of ML - Linear regression and gradient descent (continued) - Logistic regression Logistic Regression.pptx	Practical 2: - Tutorials and exercises on logistic regression and decision boundary logistic-regression-sklearn.ipynb
3 (10 th Feb.)	Lecture 3: Machine learning workflow and key issues - Typical steps in an ML workflow - Key technical issues - Ethics	Practical 3: Using logistic regression as an example to discuss and demonstrate the workflow and key issues of ML: - Tutorials and exercises on working through the workflow in an ML task using an ML algorithm, e.g. logistic regression, SVM, ensembles and ANN,

	Machine Learning Workflow and Key Issues.pptx	- Going through a workflow with sklearn: data cleaning and preparation, training, validation, evaluation, prediction logistic-regression-through-a-workflow.ipynb
4 (17 th Feb.)	Lecture 4: SVM - Basics of SVM SVM.pdf	Group project assignment HANDOUT Individual assignment HANDOUT Group Allocation Starts Practical 4: Tutorials and exercises on the basics of SVM svm-with-illustrations-basics.ipynb
5 (24 th Feb.)	Lecture 5: SVM - Kernels SVM.pdf	Group Allocation DEADLINE Practical 5: Tutorials and exercises on the kernels and applications of SVM. svm-with-illustrations-kernels.ipynb
6 (3 rd Mar.)	Lecture 6: Ensembles - Bagging (Bootstrap Aggregation) - Random Forests Ensembles.pptx	Practical 6: Tutorials and exercises on ensembles (bagging and random forests) decision-trees-bagging-and-random-forests.ipynb
7 (10 th Mar.)	Lecture 7: Ensembles - Boosting Ensembles.pptx	Practical 7: Tutorials and exercises on ensembles (voting and boosting) voting-AdaBoost.ipynb Group project proposals due (10%)
8 (17 th Mar.)	Lecture 8: Introduction to Deep Neural Networks - Artificial neural networks (ANN)	Practical 8: Tutorials and exercises on building and evaluating fully-connected deep neural networks introToDNN.pynb (on both jupyterhub and Google Colab using fully connected ANN)

	- Fully connected/dense neural networks - Activation functions - Network training - Loss functions - Applications ANN.pdf	
9 (24 th Mar.)	Lecture 9: CNNs - CNN architecture - Application examples - Brief overview of other DNNs CNN.pdf	Practical 9: Tutorial and exercises on CNN <code>introToCNNs&ImageClassification.ipynb</code> (on both jupyterhub and Google Colab using CNN) Move on to the implementation, evaluation and presentation of the group project Individual assignment HANDIN
10 (31 st Mar.)	Lecture 10: No new topics Module summary and some application examples Module Summary.pdf	Practical 10: Work on the group project with a focus applying on deep learning algorithms (e.g. CNN)
7 th -25 th Apr.	Spring Break	
11 (28 th Apr.)	Lecture 11: No new topics – Qs & As on group projects	Practical 11: Work on the group project with a focus on applying on deep learning algorithms (e.g. CNN)
12 (5 th May.)	Lecture 12: No new topic – Qs & As on group projects	Practical 12: No new topic Work on the group project Group project HAND IN